

Wednesday, Apr 17

The Multinomial Logit Model

Recall that a logistic regression model can be written as

$$\log \left[\frac{P(Y_i = 1)}{1 - P(Y_i = 1)} \right] = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}.$$

This can also be written as

$$\log(\pi_{i2}/\pi_{i1}) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik},$$

or

$$\pi_{i2}/\pi_{i1} = e^{\beta_0} e^{\beta_1 x_{i1}} \cdots e^{\beta_k x_{ik}},$$

where

$$\begin{aligned}\pi_{i2} &= P(Y_i = 1), \\ \pi_{i1} &= P(Y_i = 0).\end{aligned}$$

Here the ratio of probabilities π_{i2}/π_{i1} is the *odds* that $Y_i = 1$ rather than $Y_i = 0$. Note that odds are basically the probability of one event relative to that of another event.

Let $Y_i = 1, 2, \dots, R$ denote R categories, but not necessarily ordered in any way, and let $\pi_{i1}, \pi_{i2}, \dots, \pi_{iR}$ denote the probability of each category. The *multinomial* logistic regression model can be written as

$$\begin{aligned}\log(\pi_{i2}/\pi_{i1}) &= \beta_0^{(2)} + \beta_1^{(2)} x_{i1} + \cdots + \beta_k^{(2)} x_{ik}, \\ \log(\pi_{i3}/\pi_{i1}) &= \beta_0^{(3)} + \beta_1^{(3)} x_{i1} + \cdots + \beta_k^{(3)} x_{ik}, \\ &\vdots \\ \log(\pi_{iR}/\pi_{i1}) &= \beta_0^{(R)} + \beta_1^{(R)} x_{i1} + \cdots + \beta_k^{(R)} x_{ik},\end{aligned}$$

for a system of $R - 1$ equations. This can also be written as

$$\begin{aligned}\pi_{i2}/\pi_{i1} &= e^{\beta_0^{(2)}} e^{\beta_1^{(2)} x_{i1}} \cdots e^{\beta_k^{(2)} x_{ik}}, \\ \pi_{i3}/\pi_{i1} &= e^{\beta_0^{(3)}} e^{\beta_1^{(3)} x_{i1}} \cdots e^{\beta_k^{(3)} x_{ik}}, \\ &\vdots \\ \pi_{iR}/\pi_{i1} &= e^{\beta_0^{(R)}} e^{\beta_1^{(R)} x_{i1}} \cdots e^{\beta_k^{(R)} x_{ik}},\end{aligned}$$

so that the model relates the *odds* of categories 2 through R *relative to* the first category (often called a “baseline” or “reference” category). For example, π_{i3}/π_{i1} is the odds of the third category versus the first category. Applying the exponential function to a parameter or contrast gives an *odds ratio* that concerns the change in this odds.

Some algebra shows that the category probabilities can be written as

$$\begin{aligned}\pi_{i1} &= 1 - (\pi_{i2} + \pi_{i3} + \dots + \pi_{iR}), \\ \pi_{i2} &= \frac{e^{\eta_{i2}}}{1 + e^{\eta_{i2}} + e^{\eta_{i3}} + \dots + e^{\eta_{iR}}} \\ \pi_{i3} &= \frac{e^{\eta_{i3}}}{1 + e^{\eta_{i2}} + e^{\eta_{i3}} + \dots + e^{\eta_{iR}}} \\ &\vdots \\ \pi_{iR} &= \frac{e^{\eta_{iR}}}{1 + e^{\eta_{i2}} + e^{\eta_{i3}} + \dots + e^{\eta_{iR}}}\end{aligned}$$

where

$$\begin{aligned}\eta_{i2} &= \beta_0^{(2)} + \beta_1^{(2)}x_{i1} + \dots + \beta_k^{(2)}x_{ik}, \\ \eta_{i3} &= \beta_0^{(3)} + \beta_1^{(3)}x_{i1} + \dots + \beta_k^{(3)}x_{ik}, \\ &\vdots \\ \eta_{iR} &= \beta_0^{(R)} + \beta_1^{(R)}x_{i1} + \dots + \beta_k^{(R)}x_{ik}.\end{aligned}$$

We can write this more compactly as

$$\pi_{ic} = \frac{e^{\eta_{ic}}}{1 + \sum_{t=2}^K e^{\eta_{it}}}$$

or

$$\pi_{ic} = \frac{e^{\eta_{ic}}}{\sum_{t=1}^K e^{\eta_{it}}},$$

if we let $\eta_{i1} = 0$ since $e^0 = 1$. This is useful for computing and plotting estimated probabilities for each category of the response variable.

Example: Let's consider again the `pneumo` data from the **VGAM** package.

```
library(VGAM)
m <- vglm(cbind(normal, mild, severe) ~ exposure.time,
  family = multinomial(refLevel = "normal"), data = pneumo)
summary(m)
```

Call:

```
vglm(formula = cbind(normal, mild, severe) ~ exposure.time, family = multinomial(refLevel = "normal"),
  data = pneumo)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept):1	-4.2917	0.5214	-8.23	< 2e-16 ***
(Intercept):2	-5.0598	0.5964	-8.48	< 2e-16 ***
exposure.time:1	0.0836	0.0153	5.47	4.5e-08 ***
exposure.time:2	0.1093	0.0165	6.64	3.2e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,2]/mu[,1]), log(mu[,3]/mu[,1])

Residual deviance: 13.93 on 12 degrees of freedom

Log-likelihood: -29.54 on 12 degrees of freedom

Number of Fisher scoring iterations: 5

Warning: Hauck-Donner effect detected in the following estimate(s):
'(Intercept):1', '(Intercept):2'

Reference group is level 1 of the response

Note: The categories/levels of the response variable correspond to the order they are specified in `cbind`.

Odds ratios can be obtained in the usual way.

```
exp(cbind(coef(m), confint(m)))
```

```
                2.5 %  97.5 %  
(Intercept):1  0.013682 0.004924 0.03802  
(Intercept):2  0.006347 0.001972 0.02043  
exposure.time:1 1.087156 1.055079 1.12021  
exposure.time:2 1.115481 1.080048 1.15208
```

Here is another nice way to output the parameter estimates.

```
t(coef(m, matrix = TRUE))
```

```
                (Intercept) exposure.time  
log(mu[,2]/mu[,1])      -4.292      0.08357  
log(mu[,3]/mu[,1])      -5.060      0.10929
```

Then we can obtain odds ratio as follows.

```
exp(t(coef(m, matrix = TRUE)))
```

```
                (Intercept) exposure.time  
log(mu[,2]/mu[,1])      0.013682      1.087  
log(mu[,3]/mu[,1])      0.006347      1.115
```

Plotting the estimated category probabilities can be done as with previous models. First we create a data frame of estimated probabilities by exposure time and category.

```
d <- data.frame(exposure.time = seq(5, 52, length = 100))  
d <- cbind(d, predict(m, newdata = d, type = "response"))  
head(d)
```

```
  exposure.time normal    mild  severe  
1           5.000 0.9692 0.02014 0.01062  
2           5.475 0.9679 0.02093 0.01117  
3           5.949 0.9665 0.02174 0.01175  
4           6.424 0.9651 0.02259 0.01236  
5           6.899 0.9635 0.02346 0.01300  
6           7.374 0.9620 0.02437 0.01367
```

```
library(tidyr)  
d <- d %>% pivot_longer(cols = c(normal, mild, severe),  
  names_to = "condition", values_to = "probability")  
head(d)
```

```
# A tibble: 6 x 3
```

```
  exposure.time condition probability  
    <dbl> <chr>          <dbl>
```

1	5	normal	0.969
2	5	mild	0.0201
3	5	severe	0.0106
4	5.47	normal	0.968
5	5.47	mild	0.0209
6	5.47	severe	0.0112

Next I reorder the factor levels just for aesthetic purposes.

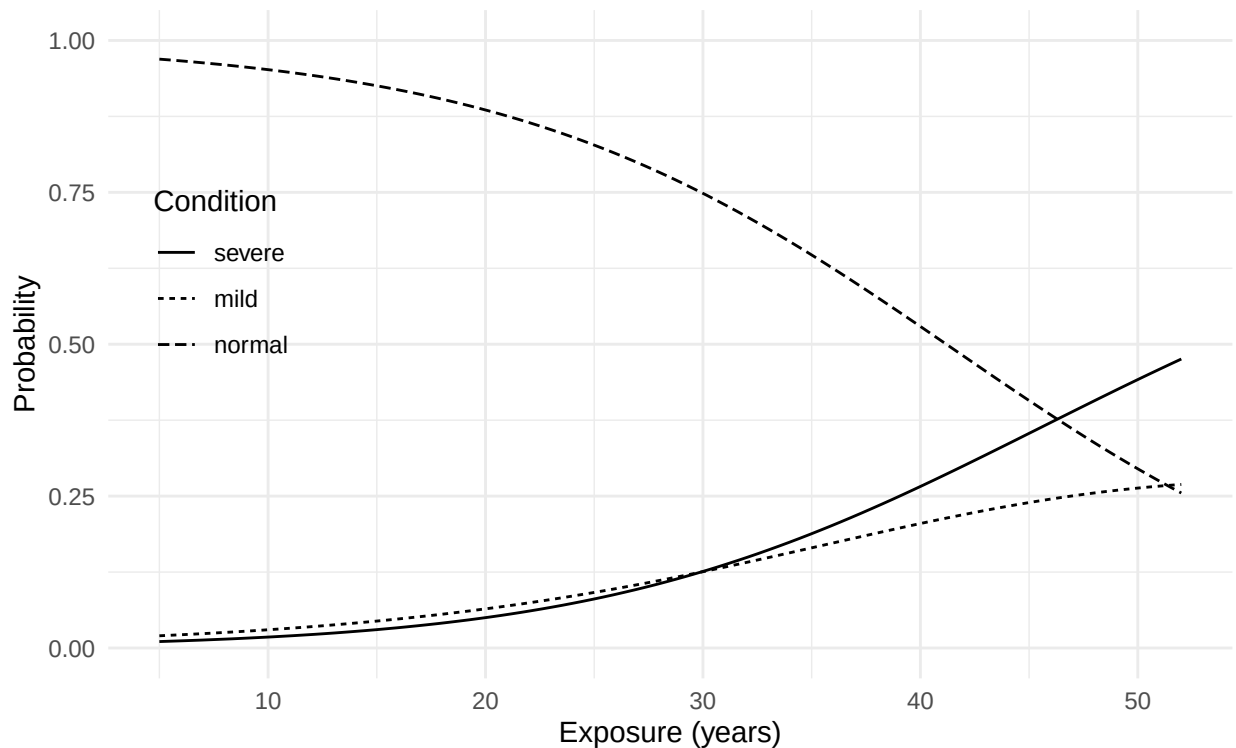
```
d$condition <- factor(d$condition, levels = c("severe","mild","normal"))
```

And then finally we plot.

```
p <- ggplot(d, aes(x = exposure.time, y = probability)) +
  geom_line(aes(linetype = condition)) +
  ylim(0, 1) + theme_minimal() + theme(legend.position = c(0.1, 0.6)) +
  labs(x = "Exposure (years)", y = "Probability", linetype = "Condition")
```

Warning: A numeric `legend.position` argument in `theme()` was deprecated in ggplot2 3.5.0. i Please use the `legend.position.inside` argument of `theme()` instead. This warning is displayed once every 8 hours. Call `lifecycle::last\_lifecycle\_warnings()` to see where this warning was generated.

```
plot(p)
```



Example: Consider the data frame `alligator` from the **EffectStars** package.

```
library(EffectStars)
data(alligator)
head(alligator)
```

	Food	Size	Gender	Lake
1	fish	<2.3	male	Hancock
2	fish	<2.3	male	Hancock

```
3 fish <2.3    male Hancock
4 fish <2.3    male Hancock
5 fish <2.3    male Hancock
6 fish <2.3    male Hancock
```

```
summary(alligator)
```

```
      Food      Size      Gender      Lake
bird  :13    <2.3:124  female: 89   George :63
fish  :94    >2.3: 95   male  :130   Hancock :55
invert:61
other :32
rep   :19
      Oklawaha:48
      Trafford:53
```

For illustration we will just consider just size and gender as explanatory variables.

```
m <- vglm(Food ~ Gender + Size, data = alligator,
  family = multinomial(refLevel = "bird"))
summary(m)
```

Call:

```
vglm(formula = Food ~ Gender + Size, family = multinomial(refLevel = "bird"),
  data = alligator)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept):1	2.0324	0.5204	3.91	9.4e-05	***
(Intercept):2	1.9897	0.5265	3.78	0.00016	***
(Intercept):3	1.1748	0.5640	2.08	0.03724	*
(Intercept):4	-0.0526	0.6829	-0.08	0.93859	
Gendermale:1	0.6149	0.6338	0.97	0.33197	
Gendermale:2	0.5247	0.6589	0.80	0.42585	
Gendermale:3	0.4185	0.7030	0.60	0.55162	
Gendermale:4	0.5833	0.7841	0.74	0.45691	
Size>2.3:1	-0.7535	0.6439	-1.17	0.24193	
Size>2.3:2	-1.6746	0.6788	-2.47	0.01362	*
Size>2.3:3	-0.9865	0.7143	-1.38	0.16723	
Size>2.3:4	0.1145	0.7962	0.14	0.88565	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,2]/mu[,1]), log(mu[,3]/mu[,1]), log(mu[,4]/mu[,1]), log(mu[,5]/mu[,1])

Residual deviance: 588.2 on 864 degrees of freedom

Log-likelihood: -294.1 on 864 degrees of freedom

Number of Fisher scoring iterations: 5

No Hauck-Donner effect found in any of the estimates

Reference group is level 1 of the response

To help interpret the output let's check the level order.

```
levels(alligator$Food)
```

```
[1] "bird" "fish" "invert" "other" "rep"
```

Extract parameter estimates and confidence intervals.

```
cbind(coef(m), confint(m))
```

```
              2.5 % 97.5 %
(Intercept):1  2.03238  1.01239  3.0524
(Intercept):2  1.98965  0.95773  3.0216
(Intercept):3  1.17478  0.06944  2.2801
(Intercept):4 -0.05261 -1.39099  1.2858
Gendermale:1   0.61487 -0.62732  1.8571
Gendermale:2   0.52467 -0.76669  1.8160
Gendermale:3   0.41850 -0.95931  1.7963
Gendermale:4   0.58333 -0.95348  2.1201
Size>2.3:1     -0.75351 -2.01561  0.5086
Size>2.3:2     -1.67459 -3.00492 -0.3443
Size>2.3:3     -0.98650 -2.38642  0.4134
Size>2.3:4      0.11450 -1.44597  1.6750
```

```
t(coef(m, matrix = TRUE))
```

```
              (Intercept) Gendermale Size>2.3
log(mu[,2]/mu[,1])      2.03238      0.6149 -0.7535
log(mu[,3]/mu[,1])      1.98965      0.5247 -1.6746
log(mu[,4]/mu[,1])      1.17478      0.4185 -0.9865
log(mu[,5]/mu[,1])     -0.05261      0.5833  0.1145
```

Compute odds ratios.

```
exp(t(coef(m, matrix = TRUE)))
```

```
              (Intercept) Gendermale Size>2.3
log(mu[,2]/mu[,1])      7.6322      1.849  0.4707
log(mu[,3]/mu[,1])      7.3130      1.690  0.1874
log(mu[,4]/mu[,1])      3.2374      1.520  0.3729
log(mu[,5]/mu[,1])      0.9487      1.792  1.1213
```

Note that we can change the reference/baseline category. This changes the model parameterization but does not change the estimated probabilities.

Joint tests of the parameters for each explanatory variable can be conducted (via a likelihood ratio test) using `anova`.

```
anova(m)
```

Analysis of Deviance Table (Type II tests)

Model: 'multinomial', 'VGAMcategorical'

Link: 'multilogitlink'

Response: Food

	Df	Deviance	Resid. Df	Resid. Dev	Pr(>Chi)
Gender	4	1.03	868	589	0.9052

```
Size      4      14.08      868      602      0.0071 **
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Note that for other models we should use `anova` by specifying a null model, but here the `anova` function does that automatically.

Here are the estimated probabilities.

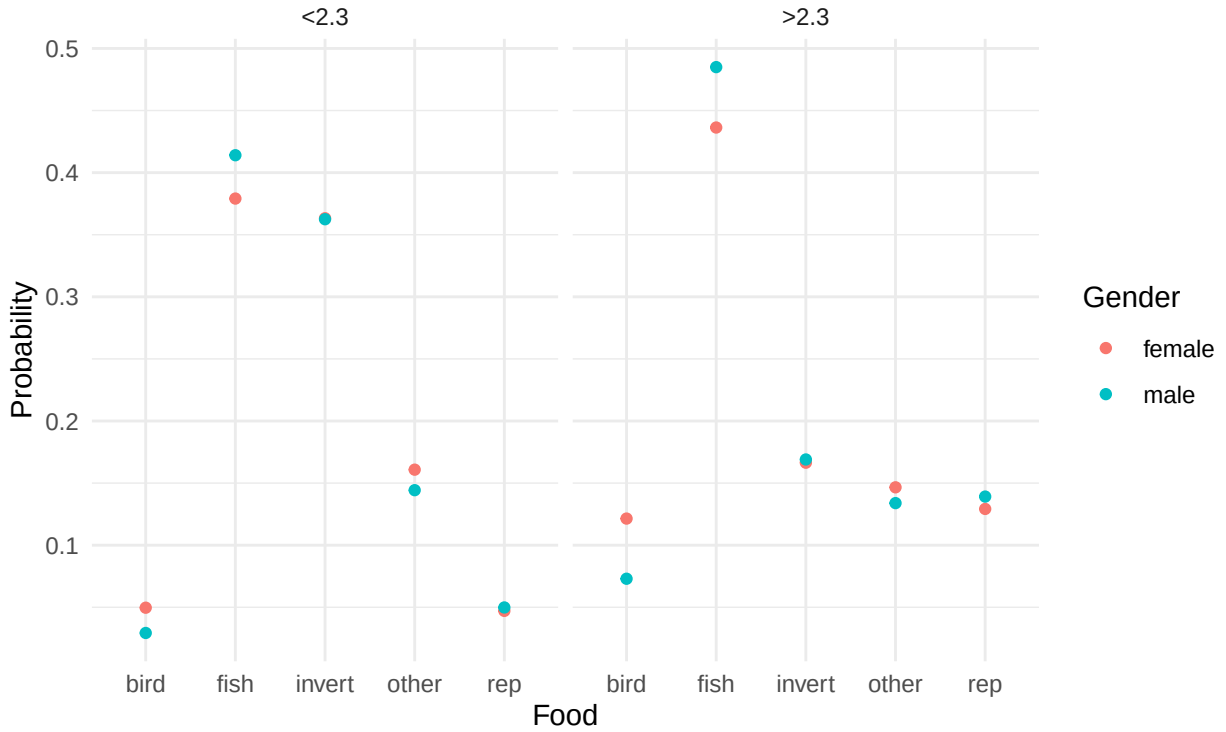
```
d <- expand.grid(Gender = c("female", "male"), Size = c("<2.3", ">2.3"))
d <- cbind(d, predict(m, newdata = d, type = "response"))
head(d)
```

```
  Gender Size  bird  fish invert  other    rep
1 female <2.3 0.04967 0.3791 0.3633 0.1608 0.04713
2  male <2.3 0.02933 0.4140 0.3625 0.1443 0.04987
3 female >2.3 0.12145 0.4363 0.1664 0.1466 0.12920
4  male >2.3 0.07299 0.4849 0.1690 0.1339 0.13914
```

```
library(tidyr)
d <- d %>% pivot_longer(cols = c(bird, fish, invert, other, rep),
  names_to = "food", values_to = "probability")
head(d)
```

```
# A tibble: 6 x 4
  Gender Size  food  probability
  <fct>  <fct> <chr>      <dbl>
1 female <2.3  bird      0.0497
2 female <2.3  fish      0.379
3 female <2.3  invert    0.363
4 female <2.3  other     0.161
5 female <2.3  rep       0.0471
6 male   <2.3  bird      0.0293
```

```
p <- ggplot(d, aes(x = food, y = probability)) + theme_minimal() +
  geom_point(aes(color = Gender)) + facet_wrap(~ Size) +
  labs(x = "Food", y = "Probability", color = "Gender")
plot(p)
```



Category-Specific Explanatory Variables

The multinomial logit model can be extended when explanatory variables vary by *response category*. For example, consider the data frame `TravelMode` from the **AER** package.

```
library(AER)
data(TravelMode)
head(TravelMode, 8)
```

	individual	mode	choice	wait	vcost	travel	gcost	income	size
1	1	air	no	69	59	100	70	35	1
2	1	train	no	34	31	372	71	35	1
3	1	bus	no	35	25	417	70	35	1
4	1	car	yes	0	10	180	30	35	1
5	2	air	no	64	58	68	68	30	2
6	2	train	no	44	31	354	84	30	2
7	2	bus	no	53	25	399	85	30	2
8	2	car	yes	0	11	255	50	30	2

Here waiting time (`wait`), vehicle cost (`vcost`), and travel time (`travel`) vary by travel mode, but household income (`income`) varies only by the respondent. For simplicity let's only consider waiting time and income as explanatory variables. A multinomial logit model can then be written as

$$\begin{aligned}\log(\pi_{ia}/\pi_{ic}) &= \beta_0^{(a)} + \beta_1(\text{wait}_i^{(a)} - \text{wait}_i^{(c)}) + \beta_2^{(a)}\text{income}_i, \\ \log(\pi_{it}/\pi_{ic}) &= \beta_0^{(t)} + \beta_1(\text{wait}_i^{(t)} - \text{wait}_i^{(c)}) + \beta_2^{(t)}\text{income}_i, \\ \log(\pi_{ib}/\pi_{ic}) &= \beta_0^{(b)} + \beta_1(\text{wait}_i^{(b)} - \text{wait}_i^{(c)}) + \beta_2^{(b)}\text{income}_i.\end{aligned}$$

If we define

$$\begin{aligned}\eta_i^{(a)} &= \beta_0^{(a)} + \beta_1(\text{wait}_i^{(a)} - \text{wait}_i^{(c)}) + \beta_2^{(a)} \text{income}_i, \\ \eta_i^{(t)} &= \beta_0^{(t)} + \beta_1(\text{wait}_i^{(t)} - \text{wait}_i^{(c)}) + \beta_2^{(t)} \text{income}_i, \\ \eta_i^{(b)} &= \beta_0^{(b)} + \beta_1(\text{wait}_i^{(b)} - \text{wait}_i^{(c)}) + \beta_2^{(b)} \text{income}_i,\end{aligned}$$

and $\eta_i^{(c)} = 0$, then we can write the category probabilities as

$$\begin{aligned}\pi_{ia} &= \frac{e^{\eta_i^{(a)}}}{e^{\eta_i^{(a)}} + e^{\eta_i^{(t)}} + e^{\eta_i^{(b)}} + e^{\eta_i^{(c)}}}, \\ \pi_{it} &= \frac{e^{\eta_i^{(t)}}}{e^{\eta_i^{(a)}} + e^{\eta_i^{(t)}} + e^{\eta_i^{(b)}} + e^{\eta_i^{(c)}}}, \\ \pi_{ib} &= \frac{e^{\eta_i^{(b)}}}{e^{\eta_i^{(a)}} + e^{\eta_i^{(t)}} + e^{\eta_i^{(b)}} + e^{\eta_i^{(c)}}}, \\ \pi_{ic} &= \frac{e^{\eta_i^{(c)}}}{e^{\eta_i^{(a)}} + e^{\eta_i^{(t)}} + e^{\eta_i^{(b)}} + e^{\eta_i^{(c)}}}.\end{aligned}$$

The quantities $e^{\eta_i^{(a)}}$, $e^{\eta_i^{(b)}}$, $e^{\eta_i^{(c)}}$, and $e^{\eta_i^{(c)}}$ could be loosely interpreted as the relative value or “utility” of each response/choice to the respondent/chooser.

Example: The `mlogit` function from the **mlogit** package will estimate a multinomial logistic regression model of this type.¹

```
library(mlogit)
m <- mlogit(choice ~ wait | income, reflevel = "car",
  alt.var = "mode", chid.var = "individual", data = TravelMode)
summary(m)
```

Call:

```
mlogit(formula = choice ~ wait | income, data = TravelMode, reflevel = "car",
  alt.var = "mode", chid.var = "individual", method = "nr")
```

Frequencies of alternatives:choice

```
car  air train  bus
0.281 0.276 0.300 0.143
```

nr method

```
5 iterations, 0h:0m:0s
g'(-H)^-1g = 0.000429
successive function values within tolerance limits
```

Coefficients :

	Estimate	Std. Error	z-value	Pr(> z)	
(Intercept):air	5.98299	0.80797	7.40	1.3e-13	***
(Intercept):train	5.49392	0.63354	8.67	< 2e-16	***
(Intercept):bus	4.10653	0.67020	6.13	8.9e-10	***
wait	-0.09773	0.01053	-9.28	< 2e-16	***
income:air	-0.00597	0.01151	-0.52	0.604	
income:train	-0.06353	0.01367	-4.65	3.4e-06	***
income:bus	-0.03002	0.01511	-1.99	0.047	*

¹This model can also be estimated using the `vglm` function from the **VGAM** package, although the syntax is very different.

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Log-Likelihood: -192
McFadden R^2: 0.322
Likelihood ratio test : chisq = 183 (p.value = <2e-16)
```

```
cbind(coef(m), confint(m))
```

		2.5 %	97.5 %
(Intercept):air	5.982989	4.39940	7.5665779
(Intercept):train	5.493920	4.25221	6.7356264
(Intercept):bus	4.106526	2.79295	5.4201038
wait	-0.097731	-0.11838	-0.0770853
income:air	-0.005967	-0.02853	0.0165931
income:train	-0.063531	-0.09033	-0.0367306
income:bus	-0.030019	-0.05964	-0.0003948

```
exp(cbind(coef(m), confint(m)))
```

		2.5 %	97.5 %
(Intercept):air	396.6241	81.4020	1932.5157
(Intercept):train	243.2087	70.2608	841.8706
(Intercept):bus	60.7354	16.3291	225.9026
wait	0.9069	0.8884	0.9258
income:air	0.9941	0.9719	1.0167
income:train	0.9384	0.9136	0.9639
income:bus	0.9704	0.9421	0.9996

Example: Here the response variable is the choice of one of three types of soda. Note that the **PoEdata** package must be installed using `remotes::install_github("https://github.com/ccolonescu/PoEdata")`.

```
library(dplyr)
library(PoEdata)
data(cola)
```

```
mycola <- cola %>% mutate(mode = rep(c("Pepsi","7-Up","Coke"), n()/3)) %>%
  select(id, mode, choice, price, feature, display) %>%
  mutate(feature = factor(feature, levels = 0:1, labels = c("no","yes"))) %>%
  mutate(display = factor(display, levels = 0:1, labels = c("no","yes")))
```

```
head(mycola)
```

	id	mode	choice	price	feature	display
1	1	Pepsi	0	1.79	no	no
2	1	7-Up	0	1.79	no	no
3	1	Coke	1	1.79	no	no
4	2	Pepsi	0	1.79	no	no
5	2	7-Up	0	1.79	no	no
6	2	Coke	1	0.89	yes	yes

```
m <- mlogit(choice ~ price + feature + display | 1, data = mycola,
  alt.var = "mode", chid.var = "id")
summary(m)
```

Call:

```
mlogit(formula = choice ~ price + feature + display | 1, data = mycola,
       alt.var = "mode", chid.var = "id", method = "nr")
```

Frequencies of alternatives:choice

```
7-Up  Coke  Pepsi
0.374 0.280 0.346
```

nr method

4 iterations, 0h:0m:0s

g'(-H)⁻¹g = 0.00174

successive function values within tolerance limits

Coefficients :

	Estimate	Std. Error	z-value	Pr(> z)
(Intercept):Coke	-0.0907	0.0640	-1.42	0.1564
(Intercept):Pepsi	0.1934	0.0620	3.12	0.0018 **
price	-1.8492	0.1887	-9.80	< 2e-16 ***
featureyes	-0.0409	0.0831	-0.49	0.6229
displayyes	0.4727	0.0935	5.05	4.3e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -1810

McFadden R²: 0.0891

Likelihood ratio test : chisq = 354 (p.value = <2e-16)

```
exp(cbind(coef(m), confint(m)))
```

		2.5 %	97.5 %
(Intercept):Coke	0.9133	0.8057	1.0353
(Intercept):Pepsi	1.2134	1.0745	1.3702
price	0.1574	0.1087	0.2278
featureyes	0.9600	0.8157	1.1297
displayyes	1.6043	1.3355	1.9271

Example: Consider the following data on choices of two options of traveling by train.

```
library(mlogit)
data(Train)
head(Train)
```

	id	choiceid	choice	price_A	time_A	change_A	comfort_A	price_B	time_B	change_B	comfort_B
1	1	1	A	2400	150	0	1	4000	150	0	1
2	1	2	A	2400	150	0	1	3200	130	0	1
3	1	3	A	2400	115	0	1	4000	115	0	0
4	1	4	B	4000	130	0	1	3200	150	0	0
5	1	5	B	2400	150	0	1	3200	150	0	0
6	1	6	B	4000	115	0	0	2400	130	0	0

There are multiple choices for each respondent (id), which can induce dependencies among the observations, but we will ignore that here. With only two choices the model reduces to logistic regression where we use the *differences* of the properties of the choices as explanatory variables.

```
m <- glm(choice == "A" ~ I(price_A - price_B) + I(time_A - time_B),
       family = binomial, data = Train)
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.018735	3.936e-02	0.476	6.341e-01
I(price_A - price_B)	-0.001024	5.939e-05	-17.237	1.400e-66
I(time_A - time_B)	-0.013968	2.288e-03	-6.106	1.021e-09

```
exp(cbind(coef(m), confint(m)))
```

	2.5 %	97.5 %
(Intercept)	1.0189	0.9433 1.1007
I(price_A - price_B)	0.9990	0.9989 0.9991
I(time_A - time_B)	0.9861	0.9817 0.9905

The price is in cents of guilders and the time is in minutes. For interpretation let's convert the scale of these variables to guilders (equal to 100 cents) and hours (equal to 60 minutes).

```
m <- glm(choice == "A" ~ I((price_A - price_B)/100) + I((time_A - time_B)/60),
        family = binomial, data = Train)
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.01874	0.039362	0.476	6.341e-01
I((price_A - price_B)/100)	-0.10236	0.005939	-17.237	1.400e-66
I((time_A - time_B)/60)	-0.83808	0.137251	-6.106	1.021e-09

```
exp(cbind(coef(m), confint(m)))
```

	2.5 %	97.5 %
(Intercept)	1.0189	0.9433 1.1007
I((price_A - price_B)/100)	0.9027	0.8921 0.9131
I((time_A - time_B)/60)	0.4325	0.3301 0.5654

Here is how we would estimate this model using `mlogit`. The data first need to be reformatted which can be done using the `dfidx` function from the `mlogit` package.

```
mytrain <- dfidx(Train, shape = "wide", choice = "choice",
                varying = 4:11, sep = "_")
head(mytrain)
```

```
~~~~~
first 10 observations out of 5858
~~~~~
  id choiceid choice price time change comfort idx
1  1         1  TRUE  2400  150      0      1 1:A
2  1         1 FALSE  4000  150      0      1 1:B
3  1         2  TRUE  2400  150      0      1 2:A
4  1         2 FALSE  3200  130      0      1 2:B
5  1         3  TRUE  2400  115      0      1 3:A
6  1         3 FALSE  4000  115      0      0 3:B
7  1         4 FALSE  4000  130      0      1 4:A
8  1         4  TRUE  3200  150      0      0 4:B
9  1         5 FALSE  2400  150      0      1 5:A
10 1         5  TRUE  3200  150      0      0 5:B

~~~ indexes ~~~
  id1 id2
1   1   A
2   1   B
3   2   A
```

```

4    2    B
5    3    A
6    3    B
7    4    A
8    4    B
9    5    A
10   5    B
indexes:  1, 2

```

```

m <- mlogit(choice ~ I(price/100) + I(time/60) | -1, data = mytrain)
summary(m)

```

Call:

```

mlogit(formula = choice ~ I(price/100) + I(time/60) | -1, data = mytrain,
        method = "nr")

```

Frequencies of alternatives:choice

```

      A      B
0.503 0.497

```

nr method

4 iterations, 0h:0m:0s

$g'(-H)^{-1}g = 1.86E-07$

gradient close to zero

Coefficients :

	Estimate	Std. Error	z-value	Pr(> z)
I(price/100)	-0.10235	0.00594	-17.2	< 2e-16 ***
I(time/60)	-0.83684	0.13722	-6.1	1.1e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -1850